Knowledge Distillation for Image Classification: A Systematic Ablation of Logit-KD and Feature-KD on CIFAR-10

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Abstract

We present a systematic ablation study of two Knowledge Distillation (KD) strategies — Logit-KD and Feature-KD — for compressing a ResNet-50 teacher into a ResNet-18 student on CIFAR-10. Using the loss formulation $L = \alpha \cdot L_{kd} + (1 - \alpha) \cdot L_{ce}$ (Hinton et al., 2015), we vary the distillation weight $\alpha \in \{0.3, 0.5, 0.7\}$ and, for Logit-KD, the softmax temperature $T \in \{2, 3, 4\}$. We conduct two experiments: Experiment 1 uses the standard torchvision ResNet architecture; Experiment 2 applies a CIFAR-specific modification (replacing the 7×7 stride-2 convolution and MaxPool with a 3×3 stride-1 convolution and Identity). This architectural fix raises teacher accuracy from 89.81% to 95.40% (+5.59pp) and enables KD to outperform the baseline student. In Experiment 2, the best Logit-KD configuration ($\alpha = 0.5, T = 4$) achieves 95.47% — +0.50pp over the 94.97% baseline. Logit-KD consistently outperforms Feature-KD across all configurations. Our results demonstrate that teacher quality and input architecture are the primary determinants of distillation effectiveness, with hyperparameter choices playing a secondary role.

1 Introduction

Deep neural networks achieve state-of-the-art performance across a wide range of computer vision tasks, but their computational demands make deployment on resource-constrained hardware challenging. Knowledge Distillation (KD) offers a principled compression strategy: instead of training a small student model on hard one-hot labels alone, the student also learns from the soft output distribution of a larger, pre-trained teacher model. These soft targets encode inter-class similarity — the relative probabilities assigned to non-target classes — a form of structured information Hinton et al. (2015) termed "dark knowledge."

Despite its widespread use, the relative effectiveness of different distillation paradigms remains context-dependent and poorly understood. Two dominant approaches exist: Logit-KD, which aligns the student's output distribution with the teacher's via KL divergence, and Feature-KD, which aligns intermediate representations at one or more layers. Both approaches introduce hyperparameters — the distillation weight α, the softmax temperature T for Logit-KD, and layer selection and alignment metrics for Feature-KD — whose optimal values depend on the task and architecture.

A key theoretical insight motivates our Logit-KD formulation: minimizing $D_{KL}(p_T \parallel p_S)$ with respect to student parameters is equivalent to maximum likelihood estimation (MLE) under the teacher's distribution, providing information-theoretic grounding for soft-target training as a principled learning objective.

In this work, we make the following contributions:

1. A systematic ablation of Logit-KD and Feature-KD on CIFAR-10, varying $\alpha \in \{0.3, 0.5, 0.7\}$ and $T \in \{2, 3, 4\}$.

2. A demonstration that standard torchvision ResNet architectures are poorly suited for 32×32 inputs, and that correcting this raises teacher accuracy by 5.59pp.
3. Evidence that teacher quality is the primary bottleneck for KD effectiveness — with a weak teacher (gap = 0.83pp), KD fails to improve over the baseline.
4. An observation that optimal temperature increases with teacher strength: $T = 2$ is best with a weak teacher, $T = 4$ with a strong teacher.

2 Related Work

2.1 Knowledge Distillation

Hinton et al. (2015) introduced KD as soft-target training, minimizing KL divergence between temperature-scaled teacher and student distributions. The combined loss $L = \alpha \cdot L_{kd} + (1 - \alpha) \cdot L_{ce}$ balances distillation against ground-truth supervision. Temperature T controls soft target sharpness: higher T exposes inter-class similarity.

2.2 Feature-Based Distillation

Romero et al. (2015) proposed FitNets, aligning intermediate feature maps via $L2$ loss after projection. Zagoruyko and Komodakis (2017) extended this to attention maps. Our Feature-KD aligns all four ResNet stages using MSE on projected features and cosine similarity on globally pooled representations.

2.3 Architecture Considerations

He et al. (2016) designed ResNets for ImageNet (224×224). For CIFAR-10 (32×32), the initial 7×7 conv (stride=2) + MaxPool (stride=2) reduces resolution to 8×8 before the first residual block — severe information loss. Standard CIFAR benchmarks replace this with a 3×3 conv (stride=1) and remove MaxPool. We adopt this modification and study its impact on distillation effectiveness.

3 Method

3.1 Loss Formulation

The total training loss follows Hinton et al. (2015):

$$L_{total} = \alpha \cdot L_{kd} + (1 - \alpha) \cdot L_{ce} \tag{1}$$

where $L_{ce} = \text{CrossEntropy}(z_S, y)$ and $\alpha \in [0, 1]$ controls the distillation weight.

3.2 Logit-KD

$$L_{logit} = T^2 \cdot D_{KL}(\text{softmax}(z_T/T) \parallel \text{softmax}(z_S/T)) \tag{2}$$

The T^2 factor compensates for reduced gradient magnitude. Minimizing $D_{KL}(p_T \parallel p_S)$ is equivalent to MLE under the teacher's distribution — the student maximizes $\mathbb{E}_{p_T}[\log p_S(x)]$, providing information-theoretic grounding for soft-target training.

3.3 Feature-KD

For each layer $i \in \{\text{layer1, layer2, layer3, layer4}\}$, we compute MSE between projected student features and teacher features, plus cosine similarity on globally averaged pooled features:

$$L_{feat} = \text{mean}_i(MSE(\text{proj}_i(S_i), T_i)) + \beta \cdot \text{mean}_i(1 - \text{cos\_sim}(GAP(S_i), GAP(T_i))) \quad (3)$$

with $\beta = 0.5$. Channel mismatches are resolved via Xavier-initialized 1×1 Conv2d projections.

3.4 CIFAR-Specific Architecture

We replace the standard conv1 (kernel=7, stride=2) and MaxPool with conv1 (kernel=3, stride=1) and Identity, applied to both teacher and student in Experiment 2. This preserves the full 32×32 spatial resolution through the first residual block.

4 Experiments

4.1 Setup

Dataset: CIFAR-10 (50k train / 10k test, 10 classes, 32×32). **Teacher:** ResNet-50 (23.5M). **Student:** ResNet-18 (11.2M). Both trained from scratch. **Augmentation:** RandomCrop(32, padding=4), RandomHorizontalFlip, Normalize. **Optimizer:** SGD (momentum=0.9, weight_decay=5×10, Nesterov). **LR:** 0.1 with CosineAnnealingLR ($T_{max} = 100, \eta_{min} = 10^{-4}$). **Batch size:** 128. **Epochs:** 100. **Hardware:** NVIDIA A100-SXM4-40GB.

4.2 Ablation Grid

- Baseline: ResNet-18 with cross-entropy only.
- Logit-KD α sweep: $\alpha \in \{0.3, 0.5, 0.7\}$, T fixed.
- Logit-KD T sweep: $T \in \{2, 3, 4\}$, $\alpha = 0.5$ (Experiment 2 only).
- Feature-KD α sweep: $\alpha \in \{0.3, 0.5, 0.7\}$, $\beta = 0.5$.

5 Results

5.1 Experiment 1 — Standard Architecture

Table 1 shows Experiment 1 results. Teacher accuracy is 89.81% and baseline student 88.98%, yielding a gap of 0.83pp. Under this condition, no KD configuration consistently outperforms the baseline. The best result, Logit-KD with $\alpha = 0.5$ and $T = 2$ (88.94%), improves only 0.04pp over baseline — within noise. Feature-KD performs worse than baseline in all configurations (up to -0.80pp).

5.2 Experiment 2 — CIFAR-Specific Architecture

Table 2 shows Experiment 2 results after the CIFAR-specific architecture modification. The teacher improves to 95.40% (+5.59pp) and baseline student to 94.97%. All Logit-KD configurations now outperform the baseline. The best result — Logit-KD with $\alpha = 0.5$ and $T = 4$ — achieves 95.47% (+0.50pp over baseline, +0.07pp over teacher). Feature-KD also improves over baseline in all configurations but remains below Logit-KD.

Table 1: Experiment 1 results (standard architecture). ✓= best KD configuration.

Config	Best Acc	Δ Baseline	Δ Teacher	Δ Exp1 Base
Teacher (ResNet-50)	89.81%	—	—	—
Baseline (ResNet-18)	88.98%	—	-0.83%	—
Logit $\alpha = 0.3, T = 4$	88.83%	-0.15%	-0.98%	—
Logit $\alpha = 0.5, T = 4$	88.88%	-0.10%	-0.93%	—
Logit $\alpha = 0.7, T = 4$	88.68%	-0.30%	-1.13%	—
Logit $\alpha = 0.5, T = 2$ ✓	88.94%	-0.04%	-0.87%	—
Logit $\alpha = 0.5, T = 8$	88.72%	-0.26%	-1.09%	—
Feature $\alpha = 0.3$	88.18%	-0.80%	-1.63%	—
Feature $\alpha = 0.5$	88.62%	-0.36%	-1.19%	—
Feature $\alpha = 0.7$	88.62%	-0.36%	-1.19%	—

Table 2: Experiment 2 results (CIFAR-specific architecture). ✓= best KD configuration.

Config	Best Acc	Δ Baseline	Δ Teacher	Δ Exp1 Base
Teacher (ResNet-50)	95.40%	—	—	+5.59%
Baseline (ResNet-18)	94.97%	—	-0.43%	+5.99%
Logit $\alpha = 0.3, T = 2$	95.37%	+0.40%	-0.03%	+6.39%
Logit $\alpha = 0.5, T = 2$	95.35%	+0.38%	-0.05%	+6.37%
Logit $\alpha = 0.7, T = 2$	95.40%	+0.43%	+0.00%	+6.42%
Logit $\alpha = 0.5, T = 3$	95.41%	+0.44%	+0.01%	+6.43%
Logit $\alpha = 0.5, T = 4$ ✓	95.47%	+0.50%	+0.07%	+6.49%
Feature $\alpha = 0.3$	95.02%	+0.05%	-0.38%	+6.04%
Feature $\alpha = 0.5$	95.01%	+0.04%	-0.39%	+6.03%
Feature $\alpha = 0.7$	95.23%	+0.26%	-0.17%	+6.25%

6 Discussion

Teacher Quality as Primary Bottleneck: The most striking finding is that teacher quality determines whether KD is beneficial at all. With a teacher-student gap of only 0.83pp (Experiment 1), KD provides no consistent improvement — soft targets carry insufficient information beyond what the student learns from hard labels. With a stronger teacher (Experiment 2, gap 0.43pp at much higher absolute accuracy), KD reliably improves the student. This suggests a non-linear relationship between gap magnitude and KD effectiveness.

Logit-KD vs. Feature-KD: Logit-KD consistently outperforms Feature-KD in both experiments. The teacher's output distribution directly encodes inter-class similarity in a task-relevant space, while feature maps encode representations that may not transfer cleanly to a smaller student architecture. Figure 1 also shows that Feature-KD convergence is smoother but slower to separate from the baseline — the α sweep curves nearly overlap.

Temperature and α Effects: Optimal temperature increases with teacher strength: $T = 2$ in Experiment 1, $T = 4$ in Experiment 2. Stronger teachers produce sharper distributions — higher T is needed to reveal inter-class similarity structure. The α sweep shows minimal sensitivity (range $\sim$0.001), suggesting distillation and ground-truth signals are largely complementary.

Architecture Dominates KD: The CIFAR-specific modification yields +5.59pp for the teacher and +5.99pp for the baseline — an order of magnitude larger than any KD gain (max +0.50pp). Input-architecture alignment is a prerequisite for effective learning, preceding distillation in the hierarchy of design decisions.

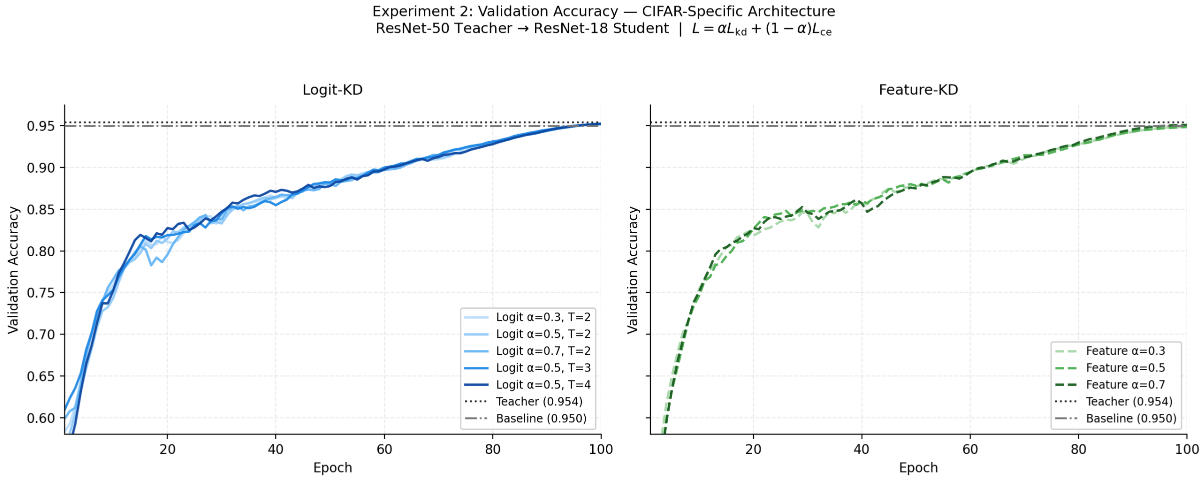

Figure 1: Validation accuracy curves for Experiment 2. Left: Logit-KD configurations (sweep and T sweep). Right: Feature-KD configurations (sweep). Dotted line = teacher (95.40%). Dash-dot line = baseline (94.97%). Curves smoothed with exponential moving average (window=9).

7 Conclusion

We presented a two-experiment ablation study of Logit-KD and Feature-KD for ResNet-50 → ResNet-18 compression on CIFAR-10. Our central finding is that teacher quality — determined primarily by architectural correctness for the input resolution — is the primary bottleneck for distillation effectiveness. When the teacher is properly trained (95.40%), Logit-KD reliably improves the student by up to +0.50pp, with $T = 4$ and $\alpha = 0.5$ as optimal hyperparameters. Feature-KD consistently underperforms Logit-KD, suggesting that task-level soft targets are more transferable than intermediate representations in this setting.

Future work will extend this investigation to more complex detection architectures (RT-DETR), where Feature-KD may better leverage spatial feature alignment, and incorporate Bayesian deep learning to study uncertainty estimation in distilled models.

References

[1] Hinton, G., Vinyals, O., & Dean, J. (2015). Distilling the Knowledge in a Neural Network. *NeurIPS Workshop on Deep Learning.*

[2] Romero, A., et al. (2015). FitNets: Hints for Thin Deep Nets. *ICLR.*

[3] Zagoruyko, S., & Komodakis, N. (2017). Paying More Attention to Attention. *ICLR.*

[4] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition. *CVPR.*

[5] Gou, J., Yu, B., Maybank, S. J., & Tao, D. (2021). Knowledge Distillation: A Survey. *IJCV*, 129(6), 1789–1819.